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Comprehensive Management of Intracranial Aneurysms Using Artificial Intelligence: An Overview

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Key words
- Artificial intelligence
- Deep learning
- Intracranial aneurysm
- Microcatheter shaping
- Outcome prediction
- Risk assessment
- Segmentation

Abbreviations and Acronyms
2D: Two-dimensional
3D: Three dimensional
3D-RA: Three dimensional rotational angiography
AAMS: AI-assisted microcatheter shaping
AI: Artificial intelligence
ANN: Artificial neural network
aSAH: Aneurysmal subarachnoid hemorrhage
AUC: Area under the receiver operating characteristic curve
CAV: Cerebral vasospasm
CNN: Convolutional neural network
CTA: Computed tomographic angiography
DCI: Delayed cerebral ischemia
DL: Deep learning
DLM: Deep learning model
DSA: Digital subtraction angiography
DSC: Dice similarity coefficient
EVT: Endovascular treatment
FCNN: Fully connected neural network
FD: Flow diverter
FP: False positive
IA: Intracranial aneurysm
ISS: In-stent stenosis
LIME: Local Interpretable Model-Agnostic Explanations
LR: Logistic regression
ML: Machine learning
MRA: Magnetic resonance angiography
RF: Random forest
SAH: Subarachnoid hemorrhage
SHAP: Shapley Additive Explanations
SVM: Support vector machine
TOF-MRA: Time-of-flight magnetic resonance angiography
UIA: Unruptured intracranial aneurysm
XGBoost: Extreme gradient boosting

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Intracranial aneurysms (IAs), an asymptomatic vascular lesion, are becoming increasingly common as imaging technology progresses. Subarachnoid hemorrhage from IAs rupture entails a substantial risk of mortality or severe disability. The early detection and prompt intervention of IAs posing a high risk of rupture are paramount for optimizing clinical management and safeguarding patients' lives. Artificial intelligence (AI), with its exceptional capabilities in image-based tasks, has garnered significant scholarly interest worldwide. Its application in the management of IAs holds promise for advancing medical research and patient care. Utilizing deep learning algorithms, AI exhibits remarkable capabilities in precisely identifying and segmenting aneurysms, significantly enhancing diagnostic sensitivity and accuracy. Furthermore, AI can meticulously analyze extensive aneurysm datasets to forecast aneurysm growth, rupture hazards, and prognostic scenarios, offering clinician's invaluable assistance in decision-making. This article comprehensively examines the latest advancements in the utilization of AI in aneurysm treatment, encompassing detection and segmentation, rupture risk assessment, prediction of therapeutic outcomes, and facilitation of microcatheter shaping. A brief discussion is held on the challenges and future paths for clinical AI deployments.

INTRODUCTION

Intracranial aneurysms (IAs), prevalent in 3.2% of the general population, are a significant medical concern owing to their life-threatening nature.[1] IAs commonly manifest as pathological dilatations of cerebral arteries due to a structural flaw in the blood vessel wall, and they predominantly occur at specific arterial bifurcations within the Circle of Willis.[2] The rupture of aneurysms is the foremost culprit behind nontraumatic subarachnoid hemorrhage (SAH), carrying a mortality rate as high as 35%[3] and resulting in a permanent disability incidence of 10%–20%.[4] Regrettably, the majority of individuals harboring asymptomatic cerebral aneurysms remain oblivious to their condition, frequently leading to a delayed diagnosis until the occurrence of aneurysm rupture, thereby complicating early intervention and management.

As noninvasive imaging methods like computed tomographic angiography (CTA) and magnetic resonance angiography (MRA) become more widely available and of higher quality; the number of incidentally discovered aneurysms is rising. On the other hand, the overwhelming workload tends to cause fatigue as the volume of scanning increases every year, thus influencing the diagnostic accuracy of IAs.[5] Early detection of aneurysms paves the way for the evaluation of rupture risk and subsequent treatment decision-making. However, regarding the necessity of intervention for an unruptured aneurysm, no agreement has been achieved as of yet.[6] Consequently, for the purpose of managing patients with aneurysms subsequently, a comprehensive assessment of the rupture risk and potential complications arising from the treatment of unruptured IAs is of utmost significance.[2]

In recent years, the utilization of artificial intelligence (AI) and deep learning (DL) in the realm of medicine has been progressively escalating. Neurosurgical preoperative lesion location and analysis, intraoperative guidance, and perioperative complications prediction have all benefited from the development of AI approaches.[7] With the assistance of DL technologies, clinicians can more precisely identify, segment, and evaluate the risk of rupture for unruptured aneurysms while also streamlining workflow.[8] Considering the management challenges of smaller unruptured aneurysms, AI is capable of autonomously providing a one-patient for a specific level of treatment approach based on extensive data training.

In this review, we will delve into the forefront of AI applications pertaining to IAs, including detection and segmentation on medical imaging, evaluation of prognosis and rupture risk, aid in the shaping of microcatheters, and post-treatment outcome prediction. We will also discuss current challenges and future prospects for AI in IA applications, such as creating a comprehensive multifunctional aneurysm management module.

CURRENT IMAGING MODALITIES FOR IAs DIAGNOSIS

The commonly utilized cerebral angiographic methodologies in clinical settings for the examination of IAs comprise CTA, MRA, and digital subtraction angiography (DSA). Each method exhibits distinct levels of precision in identifying intricate vascular abnormalities, and has been meticulously applied by medical practitioners throughout the multifaceted stages of aneurysm assessment. CTA and MRA are two widely used, noninvasive imaging modalities that facilitate the screening and expeditious diagnosis of IAs.[9] CTA is a convenient, affordable, and dependable method of screening for IAs, with a sensitivity of approximately 100% for the detection of IAs > 4mm.[10] As per the American Heart Association/American Stroke Association guidelines, CTA has been formally endorsed for the detection and subsequent monitoring of unruptured IAs (class I; level B), as well as for the work-up of aneurysmal SAH (class II b; level C).[6] The main limitations of CTA are the susceptibility to artifact and poor accuracy in detecting small IAs and those in cavernous sinus region.[11]

There are two types of MRA: direct MRA and contrast enhanced-MRA, both of which are nonradiative. Comparatively speaking, contrast enhanced-MRA is more accurate than direct MRA in showing blood vessels.[12] In comparison to CTA, MRA involves a longer scanning duration, a poorer vascular spatial resolution, a greater likelihood of missing and misdiagnosing tiny IAs, and less frequently being performed in emergency situations.[13]

Recognized as the "gold standard" for IAs detection, DSA accurately maps the spatial positioning of aneurysms owing to its unparalleled sensitivity, specificity, and image resolution. DSA's superior visualization is particularly advantageous for the accurate identification of IAs smaller than 3 mm.[10] The most recent DSA imaging method, three-dimensional rotational angiography (3D-RA), has been extensively utilized in clinical diagnosis and offers interventional doctors image information from many different perspectives.[14] But the high cost, long procedure time, possible complications, and inherent invasiveness makes it challenging to establish as a widespread disease screening technique.

AI

AI functions by constructing models and acquiring knowledge from a wide range of data repositories, subsequently manipulating this information through algorithms that mimic human cognition and reasoning capabilities. Machine learning (ML), a subset of AI, enables computers to autonomously learn and discern complex nonlinear associations within datasets through adaptive algorithms and iterative processes, thereby obviating the requirement for detailed programming instructions.[15] There are three major classifications of ML algorithms: super vised, unsupervised, and semisupervised learning or weakly supervised learning. In supervised learning, models undergo training by utilizing prelabeled datasets. Conversely, unsupervised learning operates independently of human input, analyzing unlabeled data to unveil latent relationships between variables. Semi supervised learning or weakly supervised learning utilizes a combination of prelabeled and unlabeled data for model training.[16]

DL is a specific branch within ML that employs multilayer neural networks for adaptive feature extraction and pattern recognition, with more powerful representation learning and decision-making capabilities than traditional ML. The architecture of DL network encompasses input layers, output layers, and a series of hidden layers.[17] Fully connected neural network (FCNN), recurrent neural network, and convolutional neural network (CNN) are the three primary types of typical deep learning models (DLMs). FCNN stands as the most fundamental type of neural network architecture, distinguished by the exhaustive interconnection of every node in the fully connected layer with all nodes in the superior adjacent layer. The fully connected layer, however, generally has the most parameters because all of its outputs are coupled to the inputs, necessitating a significant amount of computing and storage space.[18] Recurrent neural network is a neural network architecture specifically designed to handle sequential data.[19] CNN is excellent at deciphering pixel-level information in images and can reliably figure out pixel orientation.[20] The difference with FCNN is that the upper and lower neurons of CNN are not directly connected, but mediated by "convolutional kernels", which greatly reduces the parameters of the hidden layer through the sharing of "kernels". In contrast to traditional shallow learning-based models, CNN demonstrates minimal human intervention during the training phase and uses a back-propagation algorithm to tune the parameters in the network.[17]

BRIEF OVERVIEW OF AI MODELS DEVELOPMENT IN IAs

With respect to the detection of aneurysms from various imaging modalities, the possibility of aneurysms being undetected arises primarily due to the minuscule size and complicated structure of intracranial vessels. In addition, there may be discrepancies among interobservers about evaluating the number and location of aneurysms. Recently, the incorporation of computer-aided diagnosis systems has led to a notable improvement in diagnostic accuracy. Existing researches have mainly used image processing algorithms to detect IAs, such as region growth algorithm, fused filtering and ML methods, as well as recent DL-based techniques.[21] Since 2017, DL networks have been applied in the identification of IAs.[22,23] Nakao et al.[24] utilized a two-dimensional (2D) CNN for detection of IAs based on maximum intensity projection images, achieving 70%–94.2% sensitivity and 2.9 false positives (FPs) per case. According to Zeng et al.,[25] utilizing 2D CNN in conjunction with the spatial information fusion approach, IAs were successfully detected in unreconstructed 3D-RA images, yielding remarkable sensitivity and specificity of 99.4% and 98.2%, respectively. Another investigation led by Park et al.[26] introduced a DL-based model known as HeadXNet, possessing the capability to precisely formulate voxel-by-voxel forecasts of IAs on CTA. Unlike 2D CNN, 3D CNN exhibits superior performance by highlighting the spatial structure information of images.[27] Yang et al.[28] formulated a CNN grounded on the ResNet-18 architecture for the detection of IAs with a sensitivity of 97.5% across 649 CTA images and revealed 8 new aneurysms that had been overlooked in the first-pass diagnostic reports. Zhou et al.[29] presented a joint segmentation and classification network composed of two stages: a preliminary predetection stage, leveraging U-Net++ for a comprehensive identification of potential IAs regions, followed by a FPs reduction stage on the basis of a multipath 3D-CNN. The proposed network exhibited enhanced sensitivities and maintained a comparably lower FPs rate when compared to prior researches. Recent methods have shown sensitivities well above 95%, especially for IAs larger than 3 mm.

To our knowledge, precise segmentation serves as a crucial prerequisite for assessing the rupture risk of IAs and facilitates subsequent tailored treatment decisions. However, the accurate segmentation of IAs remains a challenging task, primarily attributed to their various shapes and intricate anatomic locations. In recent years, the exponential growth of DL has given rise to CNN-based methods that have demonstrated remarkable potential in the automatic segmentation of medical images. In ref,[30] Sichtermann et al. made a groundbreaking contribution by being the first to leverage a CNN for segmenting unruptured IAs based on a three-dimensional time-of-flight magnetic resonance angiography (3D TOF-MRA) dataset, reaching high sensitivity of 90% and dice similarity coefficient (DSC) of 0.53, respectively. Shi et al.[31] developed a DAResUNet integrated residual blocks with dual attention blocks, aiming to segment IAs from CTA images. However, the proposed model primarily concentrated on intensifying the learning of semantic information within the deeper layers, while neglecting the fine-grained details from the shallower network components. Ma and Nie[32] presented a 3D nnUNet model for training and evaluating the performance using MICCAI 2020 CADA dataset. Bo et al.[33] put forth global localization-based IA network, a fully automatic DL model designed for the segmentation of IAs on CTA, effectively integrating global localization prior and enabling the generation of detailed 3D segmentation results. Although this model offered the advantage of direct deployment across diverse clinical settings and scan configurations, eliminating the requirement for preprocessing or postprocessing, there were challenges in accurately segmenting IAs with significant size variations, particularly those of minute dimensions. Zhu et al.[34] quantitatively evaluated the influence of preprocessing and three specialized 3D segmentation models on IAs segmentation performance in scenarios with limited data availability, showcasing that 3D UNet outperformed VNet and 3D Res-UNet. In addition, Zhang et al.[35] introduced a refined neural architecture, the Dense-Dilated Neural Network, which extended the capabilities of the 3D UNet framework for the challenging task of segmenting cerebral arteries in TOF-MRA images. This network resulted in better performance than existing models, including UNet, VNet, and Uception. According to Meng et al.,[36] multimodal CNN can boost segmentation accuracy by employing multimodal processing techniques on raw brain CTA images, including Gaussian and Laplace operator. MacDonald et al.[37] presented an innovative segmentation technique with unsampled resolution and gradient enhancement for accurately capturing high curvature features like aneurysm ostia in 3D-RA images. This approach may help make the assessment of IAs morphology and rupture risk more reliable than the standard watershed method.

The further development of segmentation algorithms means the diagnosis of IAs is more accurate, which is more conductive to appropriate rupture risk assessment. Based on existing studies, the primary input variables for the intelligent prediction model are clinical, aneurysm morphological, radiomics, and hemodynamic parameters. Different algorithms are also investigated, including multivariate logistic regression (LR), support vector machine (SVM), extreme gradient boosting (XGBoost), random forest (RF), and ridge regression. Liu et al.[38] established an ML model based on PyRadiomics-derived morphological traits to predict aneurysm stability, obtained satisfactory outcomes (area under the receiver operating characteristic curve [AUC] = 0.853), and described flatness as the most important morphological predictor through lasso regression. According to Yang et al.,[39] rupture prediction is improved by integrating clinical, morphological, and hemodynamic parameters. Similarly, Turhon et al.[40] conducted an evaluation of the Transformer algorithm's performance in estimating aneurysm rupture risk in light of clinical, morphological, and radiological features. In a comparative analysis, radiomics features stood out as the most predictive, achieving an AUC of 0.738, significantly higher than the AUCs of 0.708 and 0.678 for the other two features (all $P < 0.05$). The model encompassed all features demonstrating superior performance among various combined models. Moreover, the DL model outperformed the ML and the conventional LR models, not to mention the PHASES and the UIATS score.

To date, no studies have shown that AI will be able to fully automate the treatment of IAs, while AI-assisted neurointerventional radiologists have yielded promising results in many procedures, such as cerebral angiography and intracranial aneurysm embolization. But more large prospective studies will be needed to assess the safety and generalizability.

When it comes to predicting angiographic and functional results as well as complications following aneurysm treatment, among which are microsurgical clipping and endovascular therapy, novel ML models can reach a comparatively high degree of predictive accuracy. Most recent studies have applied supervised learning algorithms, including SVM, artificial neural network (ANN), RF, Naïve Bayes, XGBoost, and CatBoost, to predict outcome in IAs after treatment.[41,42] These algorithms have proven valuable to a certain extent in assisting neurosurgeons in facilitating the informed selection of the most appropriate therapeutic modality adapted to the unique characteristics of each individual patient.

CURRENT SPECIFIC APPLICATIONS OF AI FOR IAs

The introduction of AI, especially ML and DL algorithms, has shown significant potential for streamlining the workflow in the realm of medicine. The emerging models have been applied at different phases of intracranial aneurysm management, such as detection and segmentation on medical imaging, assessment of rupture risk and prognostication, assisting to shape the microcatheters, and prediction of outcomes after treatment. The following sections provide an overview of the benefits and drawbacks of several models based on the application of AI at these stages.

AI in the Detection and Segmentation of IAs

With the continuous development of medical imaging technology, CTA, MRA, and DSA have become increasingly utilized in the screening of IAs. However, the interpretation of image results is a complex and labor-intensive process that demands precise analysis by a senior expert, especially for IAs located in unusual place and smaller than 3 mm. Recently, an increasing number of proposed algorithms and models have emerged and been applied in the detection and segmentation of IAs. The newly reported AI studies of intracranial aneurysm detection and segmentation are summarized in Table 1. Accurate detection and segmentation of IAs are a fundamental aspect in clinical decision-making, integral to the efficient management of patients harboring IAs regarding intervention and subsequent clinical outcomes.

AI Assisted Diagnosis and Segmentation on DSA or 3DRA Images. At present, DSA holds the esteemed position as the “gold standard”, attributable to its exceptional sensitivity in identifying IAs across diverse size spectra. A few cutting edge AI models have been developed to automate the detection and segmentation of IAs on DSA or 3D-RA. Podgorsak et al.[47] assessed a CNN to accurately segment both precoiling and postcoiling saccular IAs, as well as their adjacent vasculature from DSA images, with the mean DSC of 0.903 and Jaccard index of 0.823 for IAs in the testing cohort. A robust linear correlation was observed between the manually contoured masks and the automatically segmented masks generated by the network. Additionally, this study confirmed the independence of the model to the input data, regardless of precoiled and postcoiled aneurysm status. However, precoiled internal carotid artery aneurysms that are less than 3 mm in size may go unnoticed by this network because the flattening of the DSA sequences reduces their contrast with the background. Moving forward, class-activation maps[54] will be crucial for input data optimization as a means of determining which aneurysm characteristics the network considers vital.

An integrated network was designed by Ou et al.[44] that enables IAs detection, morphological analysis, and precise localization of parent artery. This network featured a novel cross-scale dual path transformer block designed to seamlessly fuse local and global information for catering to IAs of different sizes, achieving DSC of 0.832, sensitivity of 0.8, F2 score of 0.946 in the CADA test set and DSC of 0.681, sensitivity of 0.933, and F2 score of 0.909 in the external dataset. In comparison to several recent advanced algorithms, the proposed approach has demonstrated a significant enhancement in the Dice coefficient in both the validation and CADA competition dataset. However, upon evaluating on an independent database, a decline in performance was seen which is most likely related to the domain’s shift issue. Therefore, to enhance the general applicability of models, it is essential to leverage a diverse and comprehensive dataset encompassing various scanner types and originating from multiple locations.

The newest DSA imaging modality known as 3D-RA furnishes interventionalists with a multifaceted representation of anatomical structures, affording a deeper understanding of the IAs patient’s condition from diverse viewpoints.[55] Nishi et al.[46] built the first model for IAs computation without human intervention, which can efficiently extract the features of morphology from 3D-RA records. Through a hassle-free integration of a rule-driven methodology with a DL algorithm, the obtained model showed impressive accuracy in analyzing morphology and segmenting tasks. Its validation cohort data revealed DSC of 87%, testament to this method is relatively high correctness. Furthermore, this model carried out exceptionally well in estimating the highest diameter and capacity of IAs, deviating as little as possible from the defined benchmark. Unfortunately, this study is confined to a single-center context and constrained by a relatively small sample size, which may not accurately represent the real distribution of the population. Moreover, in order to reduce noise in the forecast, this model incorporates a programmed filtering mechanism that excludes lesions with volumes below one-half mm^3, which could result in the possibility of missing tiny yet real aneurysm lesions. A 3D patch-based multiclass model presented by Lin et al.[43] underwent training on a multicenter data collection and subsequent assessment on a freely accessible competition dataset. The model demonstrated proficiency in segmenting brain arteries and IAs from 3D-RA results, achieving the mean Dice value of 81%, along with the mean surface-to-surface deviation of 0.2 mm. Besides, the suggested strategy offers a promising solution to tackle the issues of class imbalance and interclass interference

Table 1. The Emerging Studies Applying Artificial Intelligence to Automatic Detection and Segmentation of Intracranial Aneurysms in Various Imaging Modalities

Type of Model	Investigator	Study Design	Sample Size: Total Cases/ Aneurysm-Positive Cases (+)/Total Aneurysms (A)	Imaging Modality	Methods	Reference Standard	Dataset: Number of Cases (Total Aneurysms [A])	Main Metrics
DSA-based models	Lin et al.[43]	Retrospective multicenter	332 cases/332+	3D-RA	DL: multiclass convolutional neural network	NR	Training: validation: testing = 7:1:2	Dice score: 0.81
	Ou et al.[44]	Retrospective multicenter	161 cases/ 161+/187A	DSA	DL: CSTB	A neurosurgeon's manual measurements	Training = 109 (127A); testing: internal = 22 (30A)/external = 30 (30A)	Internal: DSC: 0.832/ sensitivity: 0.8/precision: 0.868/F2 score: 0.946 External: DSC: 0.681/ sensitivity: 0.933/ precision: 0.823/F2 score: 0.909
	Mu et al.[45]	Retrospective single center	23 cases/23+/23A	3D-RA	DL: ARU-Net	Clinicians' manual annotation	Training = 15 (15A); testing = 8 (8A)	DSC: 0.868/sensitivity: 0.853
	Nishi et al.[46]	Retrospective single center	484 cases/ 484+/561A	3D-RA	DL: nnU-Net	A neurosurgeon and two neurointerventionalists' manual segmentation	Derivation cohort = 388 (437A); validation cohort = 96 (124A)	Dice similarity index: 0.87
	Podgorsak et al.[47]	Retrospective single center	350 cases/ 305+/313A	DSA	DL: CNN	Hand-contoured ground truth labels	Training = 250; testing = 100	DSC: 0.903/Jaccard index: 0.823
	Zhang et al.[21]	Retrospective single center	300 cases/300+	3D-RA	DL: FSTIF-UNet	5 interventional neuroadiologists' labeling	Training = 180; validation = 60; testing = 60	DSC: 0.911/HD: 1.358/ sensitivity: 0.931/IoU: 0.859
MRA-based models	Sichtermann et al.[30]	Retrospective multicenter	85 cases/ 85+/115A	TOF-MRA	DL: CNN DeepMedic	A neuroradiology resident's segmentation	Training = 58; validation = 10; testing = 17	Dice score: 0.53/ sensitivity: 0.9
	Chen et al.[48]	Retrospective multicenter	270 cases/ 270+/285A	TOF-MRA	DL: CCDU-Net	3 junior radiologists and 1 senior radiologist's manual segmentation	Training = 174; validation = 43; testing = 53 (57A)	DSC: 0.616/HD: 5.686/ VS: 0.752
	Yuan et al.[49]	Retrospective multicenter	489 cases/469+	TOF-MRA	DL: DCAU-Net	2 radiologists' manual annotation	Training = 70; validation = 20; testing: internal = 23/ external = 376 (401A)	Dice value: 0.746/ sensitivity: 0.788/ precision: 0.788
	Claux et al.[50]	Retrospective single center	49 cases/49+/63A	TOF-MRA	DL: two-stage regularized U-Net	Manual annotation	Training = 24 (38A); testing = 25 (27A)	Dice score: 0.78/ sensitivity: 0.78

3D, three dimensional; 3D-RA, three dimensional rotational angiography; +, aneurysm-positive cases; A, total aneurysms; ARU-Net, attention residual U-Net; CAL, connectivity-area-length; CNN, convolutional neural network; CSTB, cross-scale dual-path transformer module; CTA, computed tomographic angiography; DL, deep learning; DLM, deep learning model; DSA, digital subtraction angiography; DSC, dice similarity coefficient; FPs/case, false positives per case; FSTIF-UNet, full-scale temporal and spatial feature fusion-UNet; GLIA-Net, global localization-based IA network; HD, Hausdorff distance; IA, intracranial aneurysm; MRA, magnetic resonance angiography; NR, not reported; TOF-MRA, time-of-flight magnetic resonance angiography; VS, volumetric similarity; IoU, intersection over union.

Continues

Table 1. Continued

Type of Model	Investigator	Study Design	Sample Size: Total Cases/ Aneurysm-Positive Cases (+)/Total Aneurysms (A)	Imaging Modality	Methods	Reference Standard	Dataset: Number of Cases (Total Aneurysms [A])	Main Metrics
CTA-based models	Bo et al.[33]	Retrospective multicenter	1476 cases/ 1380+/1590A	CTA	DL: GLIA-Net	5 clinicians' manual annotation	Training = 1186 (1363A); testing: internal = 152 (126A)/ external A = 71 (50A)/ external B = 67 (51A)	Internal: DSC: 0.579/ recall: 0.729; External A: DSC: 0.768/recall: 0.839; External B: DSC: 0.572/ recall: 0.574
	Shahzad et al.[51]	Retrospective single center	253 cases/253+/ 294A	CTA	DL: 3D CNN (DLM-Orig, DLM-Vess and DLM-Ldim)	Radiological reports and consensus of a neurosurgeon and a radiologist	Training = 68 (79A); testing = 185 (215A)	DSC: 0.75/sensitivity: 0.82 and FPs/case: 0.81
	You et al.[20]	3190 cases/ 3190+/4124A	3190 cases/ 3190+/4124A	CTA	DL: 3D-Unet	Manual aneurysm detection and segmentation	Training = 1606 (2078A); validation = 314 (414A); testing = 352 (446A)	Dice values: 0.783/the recall: 0.964 and FPs/ case: 2.01
	Zhu et al.[34]	Retrospective single center	213 cases/101+/ 140A	CTA	DL: 3D UNet	Clinician's manual segmentation	Training= (112A); testing= (28A)	DSC: 0.818/HD: 3.323
	Patel et al.[52]	Retrospective single center	50 cases/50+	CTA and DSA	DL: CNN DeepMedic architecture	1 radiologist reviewed DSA	Training = 27; validation = 3; testing = 20	DSC: 0.868/CAL: 0.971/ precision: 0.856/recall: 0.887
	Liu et al.[53]	Retrospective multicenter	90 cases/90+/ 111A	CTA	DL: CNN DeepMedic architecture	DSA or intraoperative diagnosis	Training = 80 (98A); testing = 10 (13A)	Overall segmentation sensitivity: 0.923/ sensitivity: 0.923

3D, three dimensional; 3D-RA, three dimensional rotational angiography; +, aneurysm-positive cases; A, total aneurysms; ARU-Net, attention residual U-Net; CAL, connectivity-area-length; CNN, convolutional neural network; CSTB, cross-scale dual-path transformer module; CTA, computed tomographic angiography; DL, deep learning; DLM, deep learning model; DSA, digital subtraction angiography; DSC, dice similarity coefficient; FPs/case, false positives per case; FSTIF-UNet, full-scale temporal and spatial feature fusion-UNet; GLIA-Net, global localization-based IA network; HD, Hausdorff distance; IA, intracranial aneurysm; MRA, magnetic resonance angiography; NR, not reported; TOF-MRA, time-of-flight magnetic resonance angiography; VS, volumetric similarity; IoU, intersection over union.

problems inherent in multiclass segmentation tasks. Mu et al.[45] proposed an attention residual U-Net structure that utilizes differential preprocessing and geometric postprocessing methods to segment IAs and surrounding vessels from "patient-specific" 3D-RA data, reaching a DSC of

0.868 ± 0.038. Moreover, computational fluid dynamics simulations were performed to explore this model's clinical utility. The efficacy of depth-aware attention gates in highlighting crucial goals regarding 3D-RA images was illustrated through this study. Secondly, the employment of a multiple scales supervision technique could enable the network to harmoniously combine semantic data with spatial data across diverse hierarchical levels. In addition, this study delved into effective ways of taking away erroneous parts and bettering connected components, ultimately leading to increased segmentation precision and reproducibility.

Zhang et al.[21] built a brand-new DL-based framework termed full-scale temporal and spatial feature fusion-UNet, which was specifically tailored for self-segmentation of IAs from unreconstructed 3D-RA image sequences. The proposed model proved to have remarkable ability in merging full-scale temporal and spatial features. With the help of the Skip-Review attention method, it effectively fused long-term spatiotemporal characteristics derived from some prominent images. It also utilized a Conv-LSTM to merge the short-term spatiotemporal traits collected by several chosen 3D-RA sights taken at viewing angles that were similarly spaced. Thanks to DSC, intersection over union, and Hausdorff distance of 0.91, 0.86, and 1.36, respectively, this network's segmentation performance surpassed the models mentioned above. The suggested methodology provides a useful means of rapidly aiding neurosurgeons in the assessment of IAs.

AI Assisted Diagnosis and Segmentation on MRA Images. A number of studies have proposed AI algorithms for locating and segmenting IAs based on MRA. Chen et al.[48] adopted and trained a CCDU-Net model rooted in a coarse-to-fine segmentation architecture for segmenting unruptured intracranial aneurysms (UIAs) in multidimensional TOF-MRA. The process encompassed two pivotal components. Initially, the extraction of both vasculature and vasculature contour imagery was carried out, serving as DU-Net's dual-channel data inputs. Subsequently, a well-planned weighted loss function was developed for the training stage. Depending on the analysis of the test dataset, this DL network generated a DSC, Hausdorff distance, and volumetric similarity of 0.616, 5.946, and 0.752, respectively. Upon comparison with recent methods including nnU-Net and DeepMedic, it demonstrated superior performance. However, taking into account the actual occurrence rate of IAs in health care settings, the included patient cohort showed an unbalanced distribution and hindered the generalizability and applicability of the model. In another investigation, Yuan et al.[56] proposed a DL-based DCAU-Net, which was comprised of a convolution block attention module and dense block. For precisely and adaptively segmenting IAs on MRA pictures, this network merged both substantial semantic data from deep layers and detailed information from shallow layers, achieving the average Dice value of 0.746, sensitivity of 0.788, and precision of 0.788. On the contrary, this model is more effective for segmentation of IAs smaller than 3mm, but it needs to be further verified.

According to Claux et al.,[50] a dual-stage regularized U-Net was built employing a limited yet well-annotated database, enabling the creation of a model capable of precise IAs identification and segmentation in accordance with brain TOF-MRA. This research achieved a DSC of 75% for IAs segmentation and the lesion-wise sensitivity of 78% in the test set. However, the algorithm made relatively high false-positive detections duo to arterial bifurcation, the prior segmentation and the tortuous shape of the vessel. Beyond that, SAH tends to alter the algorithm's diagnostic performance in the setting of burst aneurysms, causing blood hyperintensity on MRA and raising the number of FPs.

AI Assisted Diagnosis and Segmentation on CTA Images. Analogous to MRA, CTA serves as an imaging tool that offers a considerably less intrusive diagnostic approach for IAs as well, featuring a comparatively high proportion of specificity and sensitivity. Along with algorithm improvement and increasing imaging resolution, the segmentation performance of CTA-based models has continuously increased. Shahzad et al.[51] trained a DLM for independently recognizing and segmenting IAs in SAH patients on CTA using a five-fold cross-validation approach. Subsequently, they developed an ensemble model (DLM-Ens) by combining the outputs of the DLM-LDim, DLM-Orig, and DLM-Vess, with a DSC of 75%, sensitivity of 0.82, and FPs per scan of 81%. Upon further scrutiny, this study found that neither the anatomical position of IAs (anterior vs. posterior circulation, with a P value of 0.07) nor the degree of bleeding (comparing Fisher grade ≤ 3 to grade 4, with a P value of 0.33) exerted a significant influence on detection and segmentation accuracy. Nevertheless, postoperative patients with IAs were omitted from this single center study, leaving an unresolved gap in our comprehension of the model's performance in such cases, given the potential complexities introduced thanks to extra artifacts. By means of a sample multicenter-based IAs dataset made up of CTA images, You et al.[20] established a 3D-Unet architecture that yielded a Dice coefficient of 78.3%, a recall of 96.4%, and FPs/case of 2.01.

AI in the Assessment of IAs Rupture Risk and Complications

When deciding on the treatment options for the asymptomatic UIAs, particularly those of smaller diameters, neurosurgeons sometimes face a challenging dilemma. They must carefully consider whether to opt for prophylactic treatment methods such as surgical clipping or endovascular management, which, albeit effective, carry inherent risks of procedure-related complications. Alternatively, they might choose conservative management, which, while avoiding immediate surgical risks, puts patients at potential risk of IAs rupture in the future. Therefore, developing a dependable system for assessing the stability of UIAs is paramount in guiding therapeutic choices. Several recent AI studies have demonstrated an AUC of 81.7%—93.6% in predicting IAs rupture status, but most of these studies have not considered explainability when developing models, which might impede their broad application in medical surroundings.[8,40,57,58] Ou et al.[59] investigated several interpretable ML-based models on a multiple-dimension dataset in order to evaluate the stability of IAs, encompassing ANN, SVM, and XGBoost. This study further demonstrated that the rationale underlying the model's predictions could be comprehensively elucidated by using Shapley Additive Explanations (SHAP) approach. In a similar study, Xiong et al.[58] harnessed the least absolute shrinkage and selection operator approach to identify forecast characteristics. Notably, among the five constructed ML prediction models, SVM emerged as the most effective, displaying an AUC value of 89.3% in external dataset. Based on the SHAP analysis, IAs characterized by thin neck, large diameter, localization in the cerebral communicating artery, and abnormal morphology were more likely to be classified as having a greater chance of rupture by the model. These findings provided physicians with valuable insights for identifying IAs posing a significant rupture risk and implementing prophylactic action accordingly. However, both Ou et al.[59] and Xiong et al.[58] only considered accessible clinical and morphological parameters. As elucidated by Mu et al.,[60] the study employed a multifaceted approach to appraise the performance and interpretability of ML methodologies. This included using the Local Interpretable Model-Agnostic Explanations (LIME) algorithm to dissect both the ensemble and separate behaviors exhibited by each method. Added to that, a permutation feature importance technique was deployed to quantitatively assess each characteristic's significance. Moreover, a local surrogate network was developed harnessing the SHAP algorithm to enable the explainability of individual predictions. Three methods claimed above were adopted to analyze six ML algorithms in this study which included morphology, anatomy, and hemodynamics variables. As shown in this research, the XGBoost model performed at its best with an AUC score of 0.78 among the assessed algorithms. Furthermore, the ML classifiers could make intelligible forecasts in line with domain-general knowledge on IAs rupture. But these methods—permutation, SHAP, and LIME algorithms, among others—were currently short of serving as unequivocal standards for a comprehensive interpretation of ML forecasts.

One of the most common clinical emergency consequences of intracranial aneurysm rupture is either cerebral vasospasm (CAV) or delayed cerebral ischemia (DCI).[61] However, AI in the prediction of aneurysm complications has not been extensively studied. Kim et al.[62] employed an interpretable modeling method to predictively explore the potential variables influencing the occurrence of CAV in aneurysmal subarachnoid hemorrhage (aSAH) patients, achieving the accuracy rate of 85.5% and AUC of 0.88. Analysis of the explainable AI system[63] proposed by Gunning showed that age, bad Fisher grade, intracranial aneurysm size, and the administration of following surgery antiplatelet drugs played pivotal roles in predicting CAV.

Drawing upon the clinical data patients at admission, Hu et al.[64] investigated a traditional LR model and ML-based networks to predict DCI after aSAH. This investigation revealed that ML algorithms demonstrated superior predictive performance and higher accuracy than traditional LR, specifically RF with an AUC of 0.858. Apart from that, an online prediction tool relying on the RF model's five predictive factors was created to accurately compute the probability rate of DCI and facilitate timely interventions. Although ML can effectively leverage input features, overfitting of data can result in subpar prediction accuracy. Taghavi et al.[65] also constructed an explainable ML system using the SHAP approach dependent on clinical factors to foresee DCI in aSAH patients. In the testing dataset, the RF algorithm obtained the highest precision in forecasting of 80.7%. SHAP analysis indicated that the top three most critical features forecasting DCI were: younger age, higher modified Fisher grade, as well as increased Hunt and Hess level. However, the results might have been impacted by the absence of certain data, particularly output data, including the relatively low modified Rankin Scale scores at the 6-month follow-up and the prevalence of DCI.

Microcatheter Shaping with AI Support for IAs Embolization

Even though microcatheter shaping is a routine technique in the interventional embolization of IAs, it can occasionally be challenging to achieve a satisfactory shape. Owing to 3D DSA technology lacking depth information, neurointerventionalists are unable to perceive the 3D cerebrovascular morphology and accurately discern the authentic trajectory of the microcatheter as it traverses the parent artery. AI algorithm can be used for appropriate microcatheter shaping which contributes to cause the tip access to aneurysmal sac stably.[66] Moreover, the therapeutic success of coil embolization is significantly influenced by the coils' ability to effectively obstruct blood flow into the aneurysmal sac. Consequently, the determination of the optimal sizes and quantity of coils to be inserted into IAs is key to ensuring a successful interventional embolization operation.[67] Liu et al.[68] conducted a preliminary study aimed at figuring out the performance of AI technology in assisting microcatheter shaping for the IAs coil embolization. In this study, AI algorithms were utilized to automate the simulation process and determine the ideal microcatheter route and mandrel form on the basis of digital imaging data of aneurysms and parent artery structures. Specifically, the collision theory, crawling algorithm, and center-line constraint were incorporated. Operators molded the microcatheter and mandrel in accordance with the AI-generated stencils, successfully executing all endovascular procedures without any perioperative complications. There was no need for reshaping and no rebound incidence. However, this study had a small number of cases and lacked a control group. Therefore, subsequent multicenter randomized controlled studies will be necessary to confirm the efficacy of this approach.

In the study conducted by Wu et al.,[69] AneuShape software functioned as a real time planning "assistant", directly integrating 3D DSA imaging data into its operational framework while performing an interventional procedure. By using collision detection algorithm and direction correction algorithm, this software could produce an optimal microcatheter shaping template. A comparative analysis revealed that the software-assisted group exhibited superior outcomes than the traditionally performed group, with increased stability, lower microcatheter reshaping rate, and improved location accuracy. Moreover, the software-assisted approach displayed significantly shorter fluoroscopy duration and reduced radiation exposure, highlighting its advantages in terms of procedural efficiency and radiation safety. Beyond that, the angiography results following aneurysm embolization showed a greater percentage of total occlusion in the software-assisted group (75 vs. 48%, $P < 0.05$).

In the meanwhile, Yang et al.[70] introduced a surgical preparation system customized for cerebral aneurysms to automatically measure multiple morphology features. Then a reverse execution of collision identification as well as centerline constraint algorithms was undertaken, commencing from the aneurysm's central point and terminating at the distal artery. This intelligently enabled the precise calculation of the microcatheter's trajectory and the generation of the shaping mandrel's geometry. The automatic output of the angle and length measured values for every segment-specific route contributed to the shaping and navigation of the microcatheter. In the AI-assisted microcatheter shaping (AAMS) group, the first-trial success rate (96 vs. 66%, $P < 0.001$), proper placement of the microcatheter under five minutes (96 vs. 72%, $P < 0.001$), and microcatheter stability (96 vs. 72%, $P < 0.05$) were significantly greater when compared with the manual shaping group. Whether for senior or junior surgeons, the AAMS group was able to effectively reduce the time required for microcatheter placement. Furthermore, this study discovered that AAMS was notably beneficial when treating cerebral aneurysm with acute inflow angle, especially if the angle was less than 60°. Nevertheless, this research shared one same flaw as those of Liu et al.[68] and Wu et al..[69] Specifically, surgeons may still be susceptible to individual prejudices when shaping microcatheter even in the presence of an AI-based template with detailed parameters and precise specification.

AI in Predicting Outcomes and Complications after IAs Intervention

Flow diverters (FDs) have emerged as a commonly utilized technique for occluding IAs while maintaining essential arterial perfusion. Since the aneurysm will not completely occlude after six months in 24% of cases, a prior evaluation model of FDs deployment results could help clinicians in treatment optimization.[71] Some investigators have treated IAs with FDs and employed aneurysm occlusion on follow-up angiography as a key outcome metric. Bhurwani et al.[72] used only angiographic parametric imaging data collected prior to and after treatment to develop a deep neural network that could foretell the occlusion outcome of IAs following therapy with the Pipeline Embolization Device. At an average follow-up of 9.8 months after treatment, occlusion prediction accuracy was noted to be 62.5% in the un-normalized cohort, rose to 70.8% in the cohort that applied arterial normalizing to angiographic parametric imaging characteristics, and further increased to 77.9% in the cohort that normalized projection views before and following treatment. Paliwal et al.[41] used 4 types of supervised ML algorithms to build predictive models, specifically LR, SVM, neural network, and k-nearest neighbor. To accurately forecast the half-yearly outcomes of IAs treated with FDs, these models were trained with a comprehensive set of sixteen variables that encompassed morphological, hemodynamic, and FDs-related properties. The findings indicated that Gaussian SVM and neural networks achieved a marginally superior accuracy of 90%, surpassing the 85% accuracy attained by k-nearest neighbor, LR, and linear SVM. However, the features employed in this investigation completely stemmed from the results of earlier researches, which might fail to entirely represent the key device, hemodynamic, and morphological characteristics in FDs-treatment. Second, the scope of this research was confined to sidewall aneurysms exclusively present in the internal carotid artery and limited to a single center. Therefore, the proposed approach might not work for FDs treating cerebral aneurysms located in other places. Unlike other studies, to predict the effect of IAs occlusion following FDs therapy, Hammoud et al.[73] constructed a ML model that was trained on a massive database containing a variety of patient risk and aneurysm morphology features. The SVM-based model suggested in this research exceeded the model by Bhurwani et al.[72] in forecasting occlusion outcomes, achieving a superior accuracy of 89% versus 77.9%. According to SHAP analysis, aneurysm neck, greater size, hypertension, age, and smoking habit were identified as key factors exerting significant influence on the accuracy of predictions. In-stent stenosis (ISS), a frequent side effect arising from the deployment of Pipeline Embolization Device, is classified based on the O'Kelly-Marotta scale.[74] This scale includes little stenosis (50%–74% luminal narrowing), serious stenosis (75%–99% luminal narrowing), and blockage (complete luminal narrowing, at 100%). Wei et al.[75] first established a predictive model for ISS by leveraging both imaging and clinical data with ML algorithms, such as elastic net, Gaussian Naïve Bayes, SVM, RF, and XGboost. Then they applied the SHAP method to give efficient elucidation for the predictions generated by the model. The elastic net exhibited the optimal predictive ability among the networks tested, as evidenced by its sensitivity of 78% and AUC of 0.71. Additionally, through SHAP analysis, it was discovered that the internal carotid artery position emerged as the leading predictive factor for ISS.

Patients with aneurysms that have undergone endovascular treatment (EVT) face a risk of recurrence of up to 33.6%.[76] On the basis of five ML algorithms: gradient boosting decision tree, deep neural network, SVM, LR, and RF classifier, Lin et al.[77] developed and evaluated prediction models designed to estimate the likelihood of recurrence among IA patients within a six-month period following EVT. The gradient boosting decision tree algorithm proved to be the top performer among the investigated models, attaining an AUC of 84.2% on the testing set. LIME and SHAP algorithms were introduced in this study to bolster both the visualization and explainability of recurrence prediction in ML models. However, extended follow-up periods are often accompanied by a heightened risk of cerebral aneurysm recurrence. A 6-month follow-up duration in this research is somewhat brief and might be deemed insufficient for thoroughly evaluating the recurrence of aneurysms following endovascular therapy. Tian et al.[78] also incorporated three ML algorithms—RF, LR, and ANNs—in building prediction models aimed at anticipating periprocedural complications arising from EVT. Out of all the models tested, the ANN obtained the best AUC of 0.76. The most significant characteristics linked to periprocedural problems, according to SHAP analysis on the ANN model, were the treatment modality and the presence of a distal aneurysm.

Surgical clipping remains an essential treatment modality acute aSAH. Using several types of AI algorithms, covering SVM, RF, generalized additive model, stochastic gradient boosting machine, generalized linear model, and decision tree, Staartjes et al.[42] established and internally validated ML models to predict functional outcomes after microsurgical clipping of IAs on a prospective registry dataset. At internal validation, the gradient boosting machine demonstrated an accuracy of 91% and an AUC of 67%. Although this model indicated potential for utilizing easily accessible preoperative data to provide patient-specific predictive analytics on complications and prognosis, additional multicenter external verification is required before being applied in clinical practice.

LIMITATIONS AND FUTURE DIRECTIONS OF AI IN MANAGING IAs

Although AI techniques show significant potential for IA management, there are still several common limitations that must be taken into consideration before being further applied in clinical practice. Most of AI studies of IAs are retrospectively designed with a relatively limited amount of data, which may carry the risk of selection bias and algorithm overfitting. These studies usually lack external validation and restrict the included population exclusively to aneurysm-positive patients for both training and testing data sets. Therefore, for the proposed models to be put into clinical use, prospective, multicenter, randomized controlled studies with standardized protocols are required for validation.

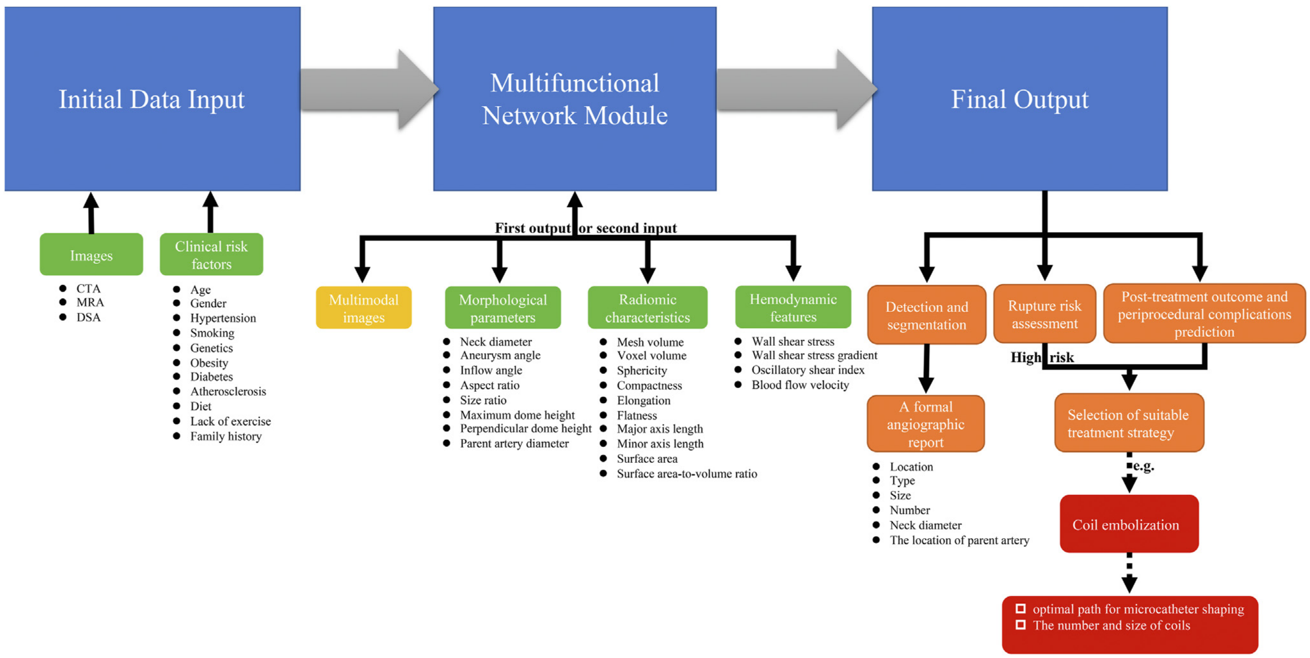

Figure 1. A hypothetical automatic multifunctional network module for full-spectrum management of intracranial aneurysms.

Traditionally, aneurysm morphology is measured manually by researchers with extensive expertise interpreting medical images. Nevertheless, manual measurement is inevitably limited by subjectivity and inconsistency, leading to discrepancies both within and among observers, as well as being unable to accurately catch the intricate geometry details of IAs.[44] This might unintentionally impact the algorithm's performance and yield aspirational outcomes that are not strictly based on reliable ground truth. A standardized double review by radiological experts in the field could partially address this challenge.

Moreover, even while recent studies have shown improved performance of AI in identifying IAs compared to previous ones, the false positive rates of these models remain significant, which hinders the clinical utility of AI. Given that most models just incorporate IA patients rather than the general population, this methodology may lead to overprediction, which accounts for the comparatively elevated incidence of FP instances noted to some extent. The vascular bifurcation, flow-related artifacts, and atherosclerosis are the most frequent sources of FPs.[79] The training process may benefit from the inclusion of both negative and positive cases. This exposes the model to variable scenarios range of scenarios that approximate real-world clinical settings, providing a reliable sensitivity and minimizing the FP rate. Additionally, training process can be conducted using a hybrid, binary cross-entropy, and DSC loss function that is skewed toward penalizing FP predictions.[52] In the aneurysm identification stage, the FPs suppression algorithm can be more effective at differentiating between IAs and vascular overlaps.[80] The pretrained U-Net++ network can efficiently boost the weights assigned to the region of interest, thus directing the multipath CNN (Cls_net) to prioritize the suspicious IAs region in the shallower network layers. This targeted approach can effectively filter out irrelevant information and substantially curtail the occurrence of FPs following predetection.[29] Adequate preprocessing and postprocessing can significantly reduce the number of FPs.

One noteworthy drawback of ML models lies in their opacity, often likened to "black boxes" that shroud the intricate relationship between input characteristics and model outputs from the scrutiny of investigators. Nevertheless, the reasoning behind the diagnosis is crucial for clinical decision making. As a consequence, it is critical to comprehend which features influence algorithm output. To conquer this inherent flaw, the SHAP analysis algorithm can be introduced to elucidate the specific contributions of individual features to the model's prediction outcomes.[58] SHAP is grounded in the principle of quantifying the average marginal contribution of each feature to the model's predicted outcomes when creating a model using several factors, consequently calculating the contribution value of each factor.[62] Hence, prior to factor selection, it is essential to evaluate the relevance of each factor to the prediction result. It is recognized that certain factors may contribute less significantly to the model's predictions. Therefore, it may be inferred that the factor is significantly correlated with the prediction if a data value is closely related to predictive outcomes. By this means, ML models can provide a predictive value for the risk of aneurysm rupture, in addition to facilitating a deeper understanding of the significance of each incorporated factor, thereby enhancing the interpretability and credibility of the model.

Another limitation is that automation bias is seen to be one of the potential pitfalls and moral dilemmas that these AI applications can bring. The term "automation bias" refers to the readers' reliance on the device and their disregard for contradictory information produced manually. Errors resulting from automation bias are likely to be further exacerbated by the usual time pressures faced by many clinicians, which will lower their faith in AI.[81] According to a systematic study, automation bias is linked to the degree of cognitive load experienced during decision-making processes. Therefore, lowering the cognitive load should be the main goal of strategies to lessen automation bias.[82]

In the coming decades, AI is expected to take on a significant role in the management of IAs. An ideal AI tool for IAs is anticipated to be a versatile network module, which can intelligently perform multiple personalized and patient-specific tasks based on readily available data, including detection and segmentation on medical imaging, evaluation of prognosis and rupture risk, choice of appropriate treatment plan, and post-treatment outcome prediction (Figure 1). The comprehensive module necessitates input parameters that are readily available and easily computed, facilitating its application in real-world clinical situations. AI-based tool for assessing cerebral vascular lesions that may also be taken into consideration in addition to aneurysms, like arteriovenous malformations, Moyamoya disease, etc. Using many advanced network topologies in tandem might potentially assist to solve these problems.[83] The AI + clinicians work mode could have a major influence on the future daily clinical practice and clinical research.

CONCLUSION

AI, especially in the realm of DL, is swiftly gaining traction as a transformative force in modern neurosurgery. Its application in the management of IAs represents promising avenues in various aspects, such as detection and segmentation in medical imaging, evaluation of prognosis and rupture risk, helping to form the microcatheter, prediction of outcomes following treatment, and selection of a suitable treatment strategy. AI-assisted microcatheter shaping emerges as a novel and ground-breaking approach in the EVT of IAs, which can assist surgeons in obtaining a precise and steady microcatheter shape rapidly. This approach is expected to gain widespread acceptance in clinical environments since it can increase the efficacy of aneurysmal occlusion, reduce radiation exposure, improve embolization density, and decrease the demand for rectification processes. Despite not yet meeting the threshold for routine clinical applications, AI provides a great deal of promise to address the patient-centered management of IAs with the use of more high quality datasets and advanced algorithms. Current research of IAs is primarily limited by small sample sizes, retrospective methods, a focus on comparison rather than prediction, and unequal data quality. Additional multicenter prospective investigations are required to further validate the applicability of these advanced algorithms in clinical practice.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Jihao Xue: Data curation, Formal analysis, Investigation, Writing – original draft. **Haowen Zheng:** Formal analysis, Writing – review & editing. **Rui Lai:** Data curation, Investigation. **Zhengjun Zhou:** Data curation, Resources. **Jie Zhou:** Project administration, Supervision. **Ligang Chen:** Funding acquisition, Supervision. **Ming Wang:** Formal analysis, Supervision, Writing – review & editing.

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